

when people want to communicate across the room. Here too, there is no reason many conversations can't occur simultaneously. By listening carefully, people can pick out individual speakers. Moreover, imagine that the people in the room could pass messages back and forth on pieces of paper. By cooperating to relay the communications, they would significantly increase the number of conversations, especially over long distances. Relaying and other cooperative techniques can serve the same function in the wireless world.

As this analogy points out, the term "interference" is problematic. Radio waves at the relevant frequencies do not bounce off one another. They pass through each other cleanly, like the intersecting ripples from two stones thrown into a pond. The overlapping signals simply make it harder for a receiver to distinguish one from another.

"Interference" is thus highly contingent on real-world factors. Again, this shouldn't be surprising. Put two television sets next to one another, and you may get a sharp picture on one but a fuzzy image on the other. The difference is that one set has a better tuner. Do we register "interference" when it shows up on one set, or both? Should the most poorly designed set define the requirements for everyone else? What if there is no set in the room but a hypothetical set with certain characteristics might experience a degraded picture there? Under current spectrum policy, such hypothetical "interference" prevents frequency sharing.

Whatever rules we set will influence behavior. If "interference" is defined with reference to a dumb receiver, vendors will try to save money and make receivers as dumb as possible. If, on the other hand, manufacturers have no guarantee of spectrum exclusivity, they will have the opposite incentive. They will build devices robust enough to deal with a variety of situations, bounded by the overall technical rules for use of the spectrum band. Those building transmitters or delivering services over the spec-